

Module 4: Visual Representation

Turning Numbers Into Pictures

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What We Will Do in This Module

- ① Make a simple bar chart
- ② Make a histogram of income
- ③ Make a box plot to compare groups
- ④ Make a scatterplot of education and income
- ⑤ Save charts for reports and papers

A good chart shows a pattern at a glance.

Why Charts Matter

- Numbers in a table can be hard to read
- A chart shows the pattern instantly
- Charts make reports and papers easier to understand

A chart is not decoration. It is a form of result representation.

The Chart Types We Will Use

For one variable

- **Bar chart** - counts of categories
- **Histogram** - distribution of a number

For two or more variables

- **Box plot** - compare groups
- **Scatterplot** - relationship between two numbers

Each chart answers a different kind of question.

Before We Begin: Load the Tools

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("village_survey.csv")

print("Tools ready and data loaded.")
```

Tools ready and data loaded.

matplotlib is the base chart tool. seaborn sits on top of it and makes prettier charts with less code.

Chart 1: A Simple Bar Chart

- A bar chart shows how many households fall into each category
- Each bar represents one category
- The height of the bar shows the count
- Good for: region, gender, land ownership, SHG membership

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Use a bar chart when one variable is a category and you want to know "how many in each group?"

Chart 1: The Code

```
sns.countplot(data=df, x='region_type')  
  
plt.title("Number of households by region")  
plt.xlabel("Region")  
plt.ylabel("Count")  
plt.show()
```

countplot counts how many rows fall into each category. Then we add a title and labels. show() displays the chart.

Chart 1: What You See

- Two bars - one for Rural, one for Urban
- The Rural bar is taller (about 130 households)
- The Urban bar is shorter (about 70 households)
- The y-axis shows the count

At a glance, we can see that most households in our dataset are rural. The chart tells us what the numbers said, but faster.

Chart 2: A Histogram

- A histogram shows how values are distributed
- Each bar represents a **range** of values
- The height shows how many households fall in that range
- Good for: income, age, wellbeing score
- Unlike a bar chart, the x-axis is a **number**, not a category

A histogram answers the question: "What does the data look like?" Is it symmetric? Skewed? Bimodal?

Chart 2: The Code

```
sns.histplot(data=df, x='income', bins=20)

plt.title("Distribution of household income")
plt.xlabel("Monthly income (INR)")
plt.ylabel("Number of households")
plt.show()
```

histplot draws the histogram. bins=20 means we divide the income range into 20 equal sections.

Chart 2: What You See

- A tall cluster of bars on the left (low incomes)
- A few short bars on the right (high incomes)
- This is called a **skewed** distribution
- Most households earn modest incomes; a few earn very high

The histogram shows what the mean could not: income is not symmetric. Most households are below the average. This is why we reported the median in Module 1.

Chart 3: A Box Plot

- A box plot compares a numeric variable across groups
- The box shows the middle 50% of values
- The line inside the box is the median
- Whiskers extend to the typical range
- Dots beyond the whiskers are outliers
- Good for: comparing income, wellbeing, or education across region or gender

A box plot shows several summary statistics at once. It is more honest than a bar chart, because it shows spread, not just the average.

Chart 3: The Code

```
sns.boxplot(data=df, x='region_type', y='income')

plt.title("Income by region type")
plt.xlabel("Region")
plt.ylabel("Monthly income (INR)")
plt.show()
```

Same structure as before. One group on the x-axis, one number on the y-axis. boxplot draws the boxes.

Chart 3: What You See

- Two boxes, one for Rural and one for Urban
- The Urban box sits higher on the y-axis
- The Urban box is also taller - more spread
- A few dots above the whiskers are high-income outliers

Not only is urban income higher on average - it is also more variable. Some urban households earn very high incomes; rural households are more similar to each other.

Chart 4: A Scatterplot

- A scatterplot shows the relationship between two numbers
- Each dot represents one household
- The x-axis shows one variable, the y-axis shows another
- Good for: education vs income, age vs wellbeing
- Helps us see whether two variables move together

A scatterplot is the visual version of correlation. If dots go up and to the right, the variables are positively correlated.

Chart 4: The Code

```
sns.scatterplot(data=df, x='education_years', y='income')

plt.title("Income vs years of education")
plt.xlabel("Years of education")
plt.ylabel("Monthly income (INR)")
plt.show()
```

Same pattern again. One variable on x, one on y. scatterplot draws the dots.

Chart 4: What You See

- Each dot is one household
- The dots drift upward from left to right
- Households with more education tend to have higher income
- A few dots sit high above the general pattern (outliers)

The scatterplot shows a positive relationship. But it is not perfect - many households break the trend. Social data always has noise.

Making Charts Look Good

- Always add a title
- Always label the x-axis and y-axis
- Include units - rupees, years, counts
- Use a clean, simple style
- Do not use 3D charts or unnecessary decorations

A chart should be readable in black and white, at a glance, by someone who has not read the report.

Saving a Chart

```
sns.boxplot(data=df, x='region_type', y='income')

plt.title("Income by region type")
plt.xlabel("Region")
plt.ylabel("Monthly income (INR)")

plt.savefig("income_by_region.png", dpi=300, bbox_inches='tight')
plt.show()
```

savefig saves the chart as an image file. dpi=300 means high quality - good for printing.
bbox_inches='tight' removes extra white space.

Saving in Different Formats

```
plt.savefig("chart.png", dpi=300)    # for slides  
plt.savefig("chart.pdf")             # for journals (vector)  
plt.savefig("chart.svg")             # for websites
```

- **PNG** - for slides and reports
- **PDF** - for journals (does not blur when zoomed)
- **SVG** - for websites (scales to any size)

Common Mistakes to Avoid

A chart is a form of writing. If a reader has to squint or guess what a bar means, the chart has failed.

- Forgetting to label the axes
- Using a bar chart when a histogram is needed
- Truncating the y-axis
- Using too many colours
- Making the chart too small to read
- Saving the chart as a screenshot instead of from code

Key Takeaways

- ① Charts show patterns faster than tables
- ② Use bar charts for categories
- ③ Use histograms for numbers
- ④ Use box plots to compare groups
- ⑤ Use scatterplots to show relationships
- ⑥ Always add titles and axis labels
- ⑦ Save charts from code, not screenshots

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Next: ethics, reproducibility, and next steps.